

Data pre-processing

- The entirety of the PhD dataset was loaded in, while only a subset of the teenager dataset was used. There are 10001 utterances for each dataset. The number of speakers varies between them, with 2700 speakers for the PhD dataset and 1699 for the teenager dataset.
- The utterances were then extracted and a column was created in each dataset to indicate whether an utterance belongs to the PhD or teenager subreddit. The two dataframes were then merged into one dataframe, then split into training and testing sets.
- The training and testing sets contain both an input and the output—since we will be predicting the subreddit from the utterance, the utterance is the input and the label is the output.
- The utterances are in text form and must be converted to a numerical form. Two methods were used for this—bag of words and TF-IDF, including an attempt to improve accuracy through n-gram features.

Vectorization

Bag of words

- The entirety of the ‘input’ for the training data is made into a list of all of the unique words inside of it
- For each utterance, a vector is created to indicate which of those unique words appear in the utterance, along with how many times each of those words appears

N-gram

- N-gram is simply bag of words but instead of the list of all unique words, it also includes unique *pairs* of words (since the n-gram range was set to (1, 2))

TF-IDF

- TF-IDF builds upon bag of words
- It keeps track of how many times each unique word appears in a given utterance, then factors in how often that word appears across the entire training dataset
- By factoring in rarity, it punishes words like 'the' and 'is' for being too common
- Bag of words would treat the word 'the' the same as the word 'thesis' if they were equally common in a given utterance

Classifier Choice

Linear SVM

- Supervised learning
- The classifier observes the training dataset, then draws a hyperplane that maximizes distance between the two clusters on either side
- To make new predictions (for testing data), it simply sees which side of the line the point falls on